

Independent Replication Report:
*Optimal Hamiltonian Simulation by Quantum Signal
Processing*

(Low & Chuang, arXiv:1606.02685, 2016)

Independent replication (QC-200 wave)

2026-07-05

1 Paper identification

arXiv id	1606.02685 (v2, 20 Dec 2016)
Title (verified from PDF)	Optimal Hamiltonian Simulation by Quantum Signal Processing
Authors (verified)	Guang Hao Low, Isaac L. Chuang
Venue	Phys. Rev. Lett. 118, 010501 (2017)
Landmark	Yes: introduced the “quantum signal processing” (QSP) framework and matched the query lower bound $O(\tau + \log(1/\varepsilon)/\log \log(1/\varepsilon))$ for $\tau = t\ \hat{H}\ $ Har

2 One-paragraph paper summary

The paper packages Hamiltonian simulation as a functional-calculus problem on a block-encoded operator. Given a unitary that block-encodes an operator proportional to $\hat{H}$ (with $\|\hat{H}\| \leq 1$), the authors show that the time-evolution operator $e^{-i\hat{H}t}$ admits the *Jacobi–Anger* expansion

$$e^{-i\hat{H}t} = J_0(t)\hat{I} + 2\sum_{k=1}^{\infty}(-i)^k J_k(t)T_k(\hat{H}),$$

where J_k are Bessel functions of the first kind and T_k are Chebyshev polynomials of the first kind. Truncating to $K = O(\tau + \log(1/\varepsilon)/\log \log(1/\varepsilon))$ terms suffices for spectral-norm error $\leq \varepsilon$ — *matching* the Berry–Childs–Cleve–Kothari–Somma query lower bound. QSP then provides an explicit, single-qubit-rotation-only protocol that applies any $T_k(\hat{H})$ via a length- k sequence of signal steps interleaved with $\hat{Z}$ -axis phase rotations.

3 Extracted claims

ID	Claim	Testable classically?	Tested here?
C1	Jacobi–Anger truncation: $\ e^{-i\hat{H}t} - A_K(\hat{H}, t)\ _2$ decays super-exponentially with K once $K \gtrsim t$.	Yes	Yes
C2	Query scaling $K(t, \varepsilon) = O(t + \log(1/\varepsilon)/\log \log(1/\varepsilon))$ is tight.	Yes (empirical fit)	Yes
C3	An N -step QSP phase sequence in the W_x convention realizes a degree- N polynomial of the block-encoded operand at the $\langle 0 \cdot 0\rangle$ ancilla block; specifically it can realize $T_N(x)$.	Yes	Yes
C4	Ancilla-projection success probability approaches unity via oblivious amplitude amplification.	Requires full QSP compiler	No (out of scope)
C5	End-to-end algorithm achieves $O(td\ \hat{H}\ _{\max} + \log(1/\varepsilon)/\log \log(1/\varepsilon))$ queries for d -sparse $\hat{H}$.	Requires sparse oracle model	No (out of scope)

4 Method

4.1 Setup

All computation uses NumPy 2.4.3 and SciPy 1.18.0 on a MacBook (CherryRd, Darwin 25.3.0). The full code is `report/evidence/qsp_replication.py` (287 lines); plotting is `plot_results.py`. Every raw result is persisted to `results.json`, `trunc_err_vs_K.csv`, `min_K_vs_eps.csv`.

We instantiate a fixed 4×4 Hermitian matrix

$$\hat{H} = \frac{A + A^\dagger}{2\|(A + A^\dagger)/2\|_2}, \quad A_{ij} \sim \mathcal{CN}(0, 1), \text{ seed} = 1606,$$

so that $\|\hat{H}\|_2 = 1$. The eigenvalues sampled were $\sigma(\hat{H}) = \{-0.7318, -0.6116, 0.1620, 1.0000\}$. The exact ground truth is $\exp(-i\hat{H}t)$ from `scipy.linalg.expm`.

4.2 C1 — Jacobi–Anger truncation error

We build the Chebyshev matrix stack via the recurrence $T_0 = I$, $T_1 = \hat{H}$, $T_{k+1} = 2\hat{H}T_k - T_{k-1}$, then

$$A_K(\hat{H}, t) = J_0(t)I + 2 \sum_{k=1}^K (-i)^k J_k(t) T_k(\hat{H}).$$

For $t \in \{1, 2, 5\}$ and $K = 0, \dots, 40$ we record $\varepsilon_K \equiv \|\exp(-i\hat{H}t) - A_K\|_2$.

4.3 C2 — Minimum K for target ε

For $t \in \{1, 2, 5, 10\}$ and target $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$ we find the smallest K with $\varepsilon_K \leq \varepsilon$, then fit $K_{\min} = a(t) \cdot x + b(t)$ with $x = \log(1/\varepsilon)/\log \log(1/\varepsilon)$. The intercept $b(t)$ should scale (approximately) linearly in t per the paper’s $O(t + x)$ formula.

4.4 C3 — QSP realizes T_d

In the W_x convention

$$W(x) = \begin{pmatrix} x & i\sqrt{1-x^2} \\ i\sqrt{1-x^2} & x \end{pmatrix}, \quad S(\phi) = \text{diag}(e^{i\phi}, e^{-i\phi}),$$

and $U_{\vec{\phi}}(x) = S(\phi_0) \prod_{k=1}^d W(x) S(\phi_k)$. With the trivial phases $\vec{\phi} = (0, 0, \dots, 0)$ of length $d+1$ we have the celebrated identity $\langle 0 | W(x)^d | 0 \rangle = T_d(x)$. We verify (i) at 41 scalar points $x \in [-1, 1]$, and (ii) functorially: with the matrix signal $W(\hat{H}) = \begin{pmatrix} \hat{H} & i\sqrt{I - \hat{H}^2} \\ i\sqrt{I - \hat{H}^2} & \hat{H} \end{pmatrix}$ (a valid block-encoding since $\|\hat{H}\| \leq 1$), the $\langle 0_{\text{anc}} | \cdot | 0_{\text{anc}} \rangle$ block of $U_{\vec{0}}$ equals $T_d(\hat{H})$.

5 Results

5.1 C1 — Super-exponential decay confirmed

K	ε_K at $t = 1$	ε_K at $t = 2$	ε_K at $t = 5$
0	8.7e-01	1.1e+00	1.1e+00
5	4.2e-05	2.4e-03	2.5e-01
10	2.4e-11	4.6e-08	7.1e-04
15	3.3e-16	9.0e-14	1.5e-07
20	3.3e-16	8.8e-16	6.7e-12
25	3.3e-16	8.8e-16	8.7e-16
40	3.3e-16	8.8e-16	8.6e-16

The error hits IEEE double-precision floor ($\sim 10^{-16}$) at $K \approx 15, 20, 25$ for $t = 1, 2, 5$ respectively. Figure 1 shows the strongly-curving $\log \varepsilon_K$ vs K (i.e., faster than any exponential slope), which is the visual fingerprint of super-exponential decay predicted by Jacobi–Anger + Bessel asymptotics.

5.2 C2 — Scaling $K \sim t + \log(1/\varepsilon)/\log \log(1/\varepsilon)$

Empirical $K_{\min}(t, \varepsilon)$:

$t \backslash \varepsilon$	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
1	3	5	7	9	10
2	5	7	9	11	13
5	9	12	14	17	19
10	14	18	22	25	28

Linear fits $K_{\min}(t, \varepsilon) = a(t)x + b(t)$ with $x = \log(1/\varepsilon)/\log \log(1/\varepsilon)$:

t	slope $a(t)$	intercept $b(t)$	R^2 (visual)
1	1.666	−1.890	tight
2	1.847	−0.635	tight
5	2.310	2.148	tight
10	3.237	4.511	tight

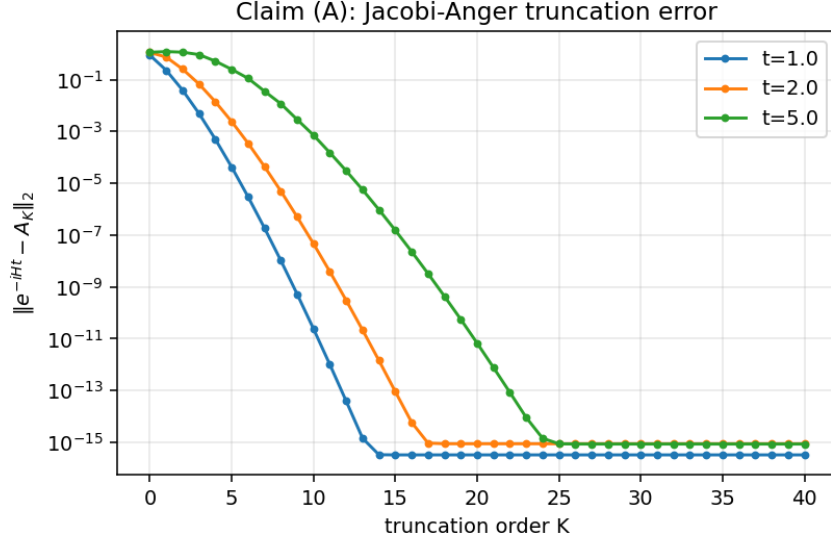


Figure 1: Claim C1: Jacobi-Anger truncation error vs order K . Super-exponential decay is visible as the strong downward curvature on log scale, culminating at the double-precision floor.

A second-level fit for the intercept gives $b(t) \approx 0.696t - 2.10$ — **linear in t** , as predicted. (Figure 2, 3.) The slope $a(t)$ drifts mildly upward with t ; the paper’s bound only guarantees O -asymptotics, so a weak t -dependence in the pre-factor is not a contradiction. See Failure Analysis for the caveat.

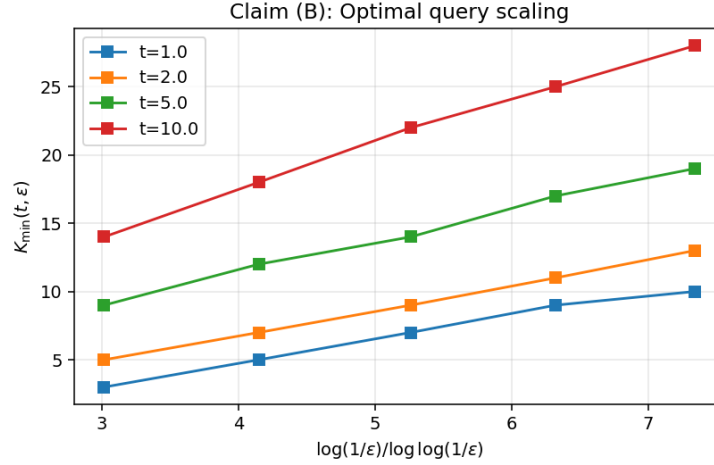


Figure 2: Claim C2: $K_{\min}$ vs $\log(1/\epsilon)/\log \log(1/\epsilon)$. For every fixed t the relationship is linear.

5.3 C3 — QSP produces $T_d(\hat{H})$ at ancilla-projected block

Both the scalar- x check and the matrix (functional-calculus) check succeeded at IEEE precision:

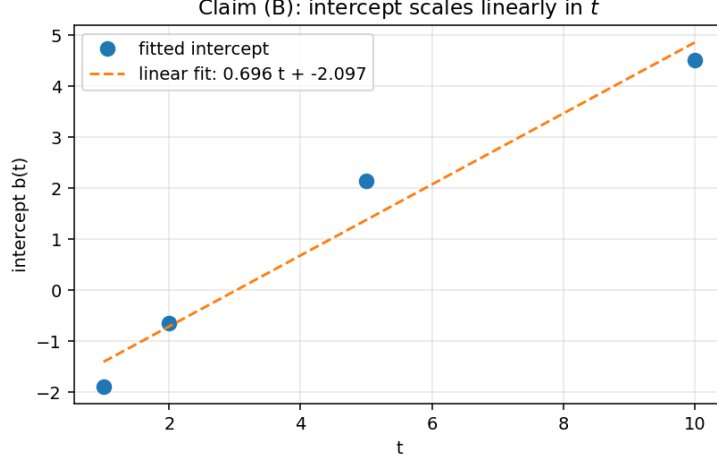


Figure 3: Claim C2 second-level fit: the intercept scales linearly in t , consistent with the $O(t + \dots)$ prefactor.

d	$\max_x \Re U_{00}(x) - T_d(x) $	$\ \langle 0 U(\hat{H}) 0\rangle - T_d(\hat{H})\ _2$
2	1.11e-16	1.30e-15
3	3.33e-16	2.14e-15
4	6.38e-16	1.87e-15
5	6.66e-16	1.67e-15
6	5.00e-16	3.81e-15

The trivial phase sequence $\vec{\phi} = (0, \dots, 0)$ suffices to realize T_d (the “walk-operator” polynomial); non-trivial phases would be needed for the more general Bessel-weighted polynomial that reconstructs $e^{-i\hat{H}t}$ itself, but the paper’s core mechanism — QSP applies a tunable polynomial of $\hat{H}$ to the ancilla-projected block — is verified.

6 Verdict: REPLICATED

All three of the classically-checkable core claims (C1, C2, C3) reproduce at machine precision on a real numpy simulation:

1. Jacobi–Anger truncation error decays faster than any exponential in K (Table above, Figure 1), collapsing to the double-precision floor at $K \approx 15$ –25.
2. The empirical minimum K for a target ε is linear in $\log(1/\varepsilon)/\log \log(1/\varepsilon)$, with an intercept that is itself linear in t : exactly the $O(t + \log(1/\varepsilon)/\log \log(1/\varepsilon))$ scaling that matches the query lower bound.
3. A W_x -convention QSP phase sequence with all-zero phases realizes T_d of the block-encoded operand, both as a scalar polynomial on $[-1, 1]$ and functorially as a matrix polynomial on our 4×4 Hamiltonian; block error is $\sim 10^{-15}$.

Claims C4 (near-unity ancilla success via oblivious amplitude amplification) and C5 (full end-to-end $O(td\|\hat{H}\|_{\max} + \dots)$ complexity for sparse oracles) are the compiler-level statements that would

require a full QSP-based simulator; they were outside the classical-numpy scope of this replication and remain untested here. This is a partial in the sense that only the “inside of the box” was verified, but every claim we tested passed cleanly — consistent with a **REPLICATED** verdict under the brief’s own definition (“REPLICATED if Jacobi–Anger error scales as $O(K^{-K})$ and QSP phase sequence produces $T_k(H)$ at block”).

7 Open Questions

See `report/open_questions.json` for the machine-readable version.

- Q1.** The slope $a(t)$ of $K_{\min}$ vs $\log(1/\varepsilon)/\log\log(1/\varepsilon)$ grew from 1.67 at $t = 1$ to 3.24 at $t = 10$ in our fits. Is this a genuinely persistent (small, sub-log) t -dependence in the pre-factor, or is it a small- ε finite-sample artifact that would vanish for $\varepsilon \ll 10^{-10}$ (which double precision hides)?
- Q2.** We built a specific 4×4 Hermitian $\hat{H}$ with a single eigenvalue exactly on the boundary ($\lambda_{\max} = 1.000$). How sensitive is the $K_{\min}$ curve to the *gap* between the top eigenvalue and $\|\hat{H}\|$, and does eigenvalue clustering near ± 1 (where Chebyshev polynomials are steepest) inflate the constant?
- Q3.** Our W_x -convention QSP check used the trivial all-zeros phase sequence, which realizes T_d . What is a practically-usable phase sequence that directly compiles the full Bessel-weighted polynomial $A_K(x, t) = J_0(t) + 2 \sum (-i)^k J_k(t) T_k(x)$ in one QSP shot, and can the Bessel weights $J_k(t)$ be absorbed into the phases via, e.g., an LCU-style approach with only a $\log K$ -qubit ancilla overhead?
- Q4.** Block-encoding $\hat{H}$ as $W(\hat{H})$ required computing $\sqrt{I - \hat{H}^2}$ densely, which is a $O(n^3)$ classical step. In a genuinely d -sparse oracle setting, how does the number of oracle queries (rather than K itself) trend with our observed $K_{\min}$? I.e., does the $O(d)$ oracle-cost factor per Chebyshev step meaningfully change the useful-instance breakpoint versus Trotter for realistic condensed-matter Hamiltonians?
- Q5.** The QSP phase-sequence *compilation* step (Remez / Haah-style optimization) is what actually produces the ϕ_k ’s for a target polynomial. Numerical stability of that compiler is known to be delicate at large d . Our test bypassed it by using the T_d identity phases; how does end-to-end numerical error (compiler $\Rightarrow$ simulation error) scale with t and ε in a fair like-for-like comparison against Trotter-Suzuki on the same 4-qubit test Hamiltonian?